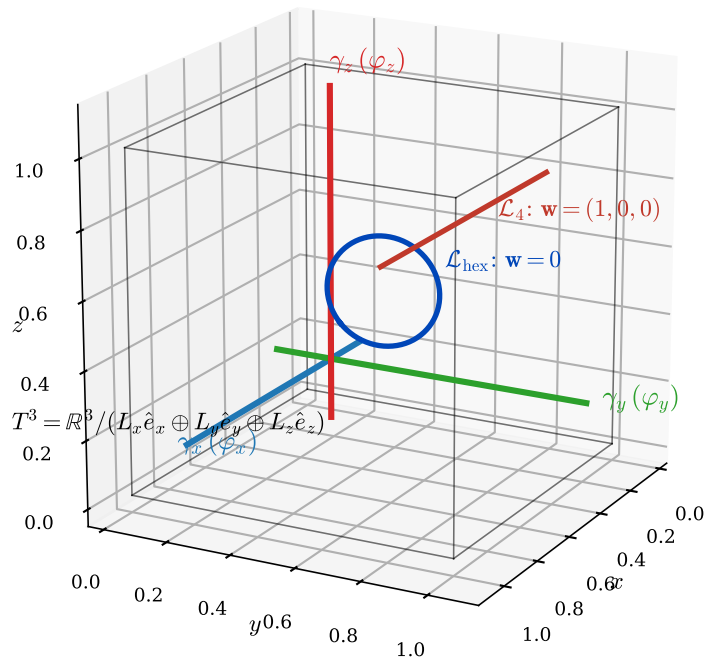
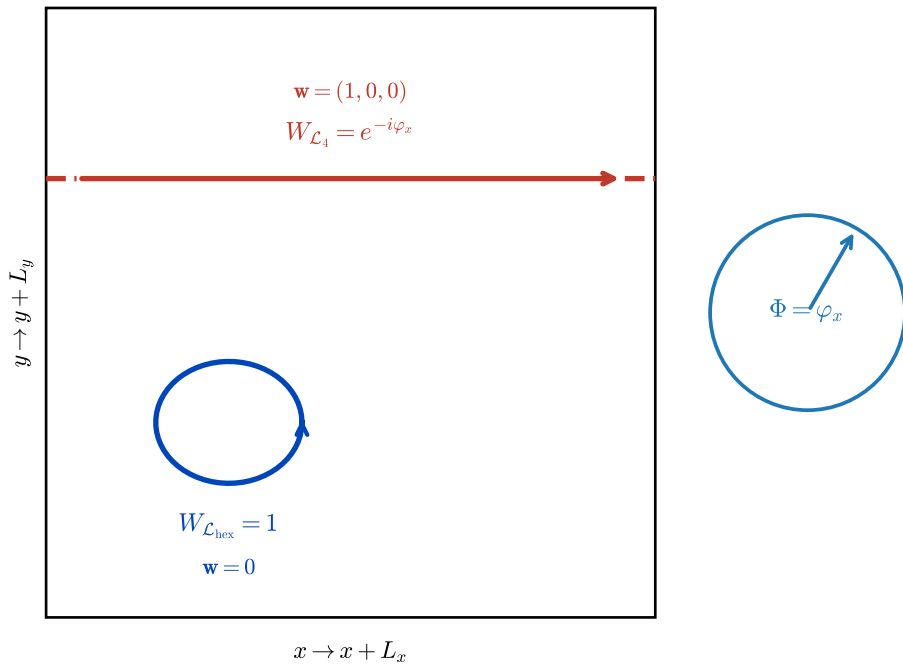


(a) Cluster as a flat 3-torus T^3 with $H_1 = \mathbb{Z}^3$



(b) Twist as Aharonov–Bohm flux through γ_x

Wilson loop: $W_{\mathcal{L}}(\varphi) = e^{-i\mathbf{w}_{\mathcal{L}} \cdot \varphi}$



Twist averaging: $\overline{W_{\mathcal{L}}} = \delta_{\mathbf{w}, 0}$